

Chemical Nomenclature – Naming Compounds the IUPAC Way

Water. Salt. Ammonia. Baking soda. Vinegar. You know these substances by their common names, but chemists around the world need a universal system to name them. That is why the International Union of Pure and Applied Chemistry (IUPAC) developed rules for naming chemical compounds. Yet many students memorize names without understanding the logic behind them.

IUPAC nomenclature allows scientists from any country to communicate clearly about chemicals. Understanding chemical nomenclature means understanding how to name compounds correctly and how to write formulas from names – skills essential for every chemist.

This reading material explores the IUPAC rules for naming common compounds, provides practice with naming specific compounds (including H_2O , NaCl , NH_3 , NaHCO_3 , H_2SO_4 , ZnSO_4 , MgO , Fe_3O_4 , CaO , and $\text{Ca}(\text{OH})_2$), and explains how to write chemical formulas for compounds commonly found in the home (including sodium chloride, ammonia, sodium fluoride, hydrochloric acid, ethanol, sucrose, acetone, and acetic acid).

Part 1: What Is IUPAC Nomenclature and Why Does It Matter?

Definition of IUPAC Nomenclature

IUPAC nomenclature is the standardized system for naming chemical compounds, established by the International Union of Pure and Applied Chemistry. It ensures that every chemical compound has a unique, universally recognized name.

The Central Idea: One Compound, One Name

Before IUPAC, the same compound could have many different names depending on the country, the scientist, or even personal preference. IUPAC created rules so that a chemist in Japan, Brazil, and Germany all use the same name for the same compound.

Why IUPAC Nomenclature Matters

Aspect	Explanation
Universal communication	Scientists worldwide use the same names for the same compounds
Unambiguous identification	Each IUPAC name corresponds to exactly one compound
Predictability	The name of a compound tells you something about its structure and composition
Safety	Correct naming prevents dangerous mistakes in laboratories and industries
Legal and regulatory	Chemical regulations and safety data sheets rely on standardized names

Part 2: Types of Chemical Compounds and Basic Naming Rules

Before naming specific compounds, you must understand the two main categories of inorganic compounds: ionic compounds and molecular (covalent) compounds. An ionic compound is formed by the transfer of electrons between a metal and a nonmetal, resulting in positive and negative ions held together by electrostatic forces. In contrast, a molecular (covalent) compound

is formed by the sharing of electrons between two or more nonmetal atoms, resulting in neutral molecules held together by covalent bonds.

Ionic Compounds vs. Molecular Compounds

Feature	Ionic Compounds	Molecular (Covalent) Compounds
Bond type	Transfer of electrons (metal + nonmetal)	Sharing of electrons (nonmetal + nonmetal)
Components	Metal + nonmetal (or polyatomic ion)	Two or more nonmetals
Naming rule	Name of metal + name of nonmetal with -ide ending	Prefixes indicate number of atoms + element names
Example	NaCl (sodium chloride)	CO ₂ (carbon dioxide)

Basic Naming Rules for Ionic Compounds

Rule	Explanation	Example
1	Write the name of the metal first (positive ion)	NaCl → sodium
2	Write the name of the nonmetal second, change the ending to -ide	NaCl → chloride
3	Combine: metal name + nonmetal name with -ide	NaCl → sodium chloride

Common -ide Endings for Nonmetals

Nonmetal	Element Name	Ion Name (in compound)
Cl	Chlorine	Chloride
O	Oxygen	Oxide
N	Nitrogen	Nitride
S	Sulfur	Sulfide
F	Fluorine	Fluoride
Br	Bromine	Bromide
I	Iodine	Iodide
P	Phosphorus	Phosphide

Basic Naming Rules for Molecular Compounds (Prefixes)

Prefix	Number	Example
mono-	1	carbon monoxide (CO)
di-	2	carbon dioxide (CO ₂)
tri-	3	nitrogen triiodide (NI ₃)
tetra-	4	carbon tetrachloride (CCl ₄)
penta-	5	dinitrogen pentoxide (N ₂ O ₅)
hexa-	6	sulfur hexafluoride (SF ₆)

Note: The prefix "mono-" is often omitted for the first element (e.g., CO is carbon monoxide, not monocarbon monoxide).

Naming Compounds with Transition Metals (Stock System)

Some metals (especially transition metals) can form ions with different charges. Roman numerals indicate the charge.

Formula	Metal Ion Charge	Name
FeO	Iron (II)	Iron(II) oxide
Fe ₂ O ₃	Iron (III)	Iron(III) oxide
CuCl	Copper (I)	Copper(I) chloride
CuCl ₂	Copper (II)	Copper(II) chloride

Part 3: Naming Common Compounds (IUPAC Rules)

Below are the IUPAC names for the ten common compounds listed in the learning objective. Each entry includes the chemical formula, the IUPAC name, the compound type, and an explanation of the naming rules applied.

Compound 1: H₂O

Property	Information
Chemical formula	H ₂ O
IUPAC name	Water (or Dihydrogen monoxide)

Property	Information
Compound type	Molecular (covalent)
Explanation	While "dihydrogen monoxide" follows the prefix rules (di = 2, mono = 1), IUPAC accepts the common name "water" due to universal usage.

Compound 2: NaCl

Property	Information
Chemical formula	NaCl
IUPAC name	Sodium chloride
Compound type	Ionic (metal + nonmetal)
Explanation	Sodium (Na^+) is the metal name; chlorine becomes chloride (nonmetal with -ide ending).

Compound 3: NH_3

Property	Information
Chemical formula	NH_3
IUPAC name	Ammonia (or Nitrogen trihydride)
Compound type	Molecular (covalent)

Explanation	While "nitrogen trihydride" follows prefix rules, IUPAC accepts the common name "ammonia" due to universal usage.
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Compound 4: NaHCO₃

Property	Information
Chemical formula	NaHCO ₃
IUPAC name	Sodium hydrogen carbonate
Compound type	Ionic (contains polyatomic ion)
Explanation	Sodium (Na ⁺) + hydrogen carbonate (HCO ₃ ⁻ , a polyatomic ion). Also commonly called sodium bicarbonate or baking soda.

Compound 5: H₂SO₄

Property	Information
Chemical formula	H ₂ SO ₄
IUPAC name	Sulfuric acid
Compound type	Acid (contains hydrogen and a polyatomic ion)
Explanation	Acids with polyatomic ions ending in -ate become -ic acid. Sulfate (SO ₄ ²⁻) → sulfuric acid.

Compound 6: ZnSO₄

Property	Information
Chemical formula	ZnSO ₄
IUPAC name	Zinc sulfate
Compound type	Ionic (contains polyatomic ion)
Explanation	Zinc (Zn ²⁺) + sulfate (SO ₄ ²⁻). Zinc has only one common charge (+2), so no Roman numeral is needed.

Compound 7: MgO

Property	Information
Chemical formula	MgO
IUPAC name	Magnesium oxide
Compound type	Ionic (metal + nonmetal)
Explanation	Magnesium (Mg ²⁺) + oxygen becomes oxide. Charges balance: Mg ²⁺ and O ²⁻ form a 1:1 ratio.

Compound 8: Fe₃O₄

Property	Information
Chemical formula	Fe ₃ O ₄
IUPAC name	Iron(II,III) oxide or Magnetite
Compound type	Ionic (transition metal with mixed oxidation states)
Explanation	Fe ₃ O ₄ contains both Fe ²⁺ and Fe ³⁺ ions. It is often called iron(II,III) oxide or by its mineral name, magnetite.

Compound 9: CaO

Property	Information
Chemical formula	CaO
IUPAC name	Calcium oxide
Compound type	Ionic (metal + nonmetal)
Explanation	Calcium (Ca ²⁺) + oxygen becomes oxide (O ²⁻). Common name: quicklime.

Compound 10: Ca(OH)₂

Property	Information
Chemical formula	Ca(OH) ₂

Property	Information
IUPAC name	Calcium hydroxide
Compound type	Ionic (contains polyatomic ion)
Explanation	Calcium (Ca^{2+}) + hydroxide (OH^- , polyatomic ion). Two hydroxide ions balance one calcium ion. Common name: slaked lime or lime water.

Part 4: Writing Chemical Formulas for Common Household Compounds

The ability to write chemical formulas from compound names is just as important as naming. Below are common compounds found in the home, with their chemical formulas and explanations.

Household Compound 1: Sodium Chloride

Property	Information
Common name	Table salt
Chemical formula	NaCl
Ions involved	Na^+ (sodium ion) and Cl^- (chloride ion)
Explanation	One sodium ion (+1) balances one chloride ion (-1). Found in every kitchen.

Household Compound 2: Ammonia

Property	Information
Common name	Household ammonia
Chemical formula	NH_3
Compound type	Molecular compound
Explanation	One nitrogen atom bonded to three hydrogen atoms. Used in glass cleaners and fertilizers.

Household Compound 3: Sodium Fluoride

Property	Information
Common name	Fluoride (in toothpaste)
Chemical formula	NaF
Ions involved	Na^+ (sodium ion) and F^- (fluoride ion)
Explanation	Added to toothpaste and drinking water to prevent tooth decay.

Household Compound 4: Hydrochloric Acid

Property	Information
Common name	Muriatic acid

Chemical formula	HCl
Compound type	Binary acid (hydrogen + halogen)
Explanation	Found in some household cleaners and concrete etchings. Also present in stomach acid (gastric juice).

Household Compound 5: Ethanol

Property	Information
Common name	Drinking alcohol, ethyl alcohol
Chemical formula	$\text{C}_2\text{H}_5\text{OH}$ or $\text{CH}_3\text{CH}_2\text{OH}$
Compound type	Organic compound (alcohol)
Explanation	The active ingredient in alcoholic beverages; also used as a disinfectant and solvent.

Household Compound 6: Sucrose

Property	Information
Common name	Table sugar
Chemical formula	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$
Compound type	Organic compound (carbohydrate, disaccharide)

Property	Information
Explanation	Extracted from sugar cane or sugar beets; used as a sweetener in cooking and baking.

Household Compound 7: Acetone

Property	Information
Common name	Nail polish remover
Chemical formula	$(\text{CH}_3)_2\text{CO}$ or $\text{C}_3\text{H}_6\text{O}$
Compound type	Organic compound (ketone)
Explanation	A colorless, flammable liquid used to remove nail polish and as a solvent.

Household Compound 8: Acetic Acid

Property	Information
Common name	Vinegar (dilute solution)
Chemical formula	CH_3COOH or $\text{C}_2\text{H}_4\text{O}_2$
Compound type	Organic compound (carboxylic acid)
Explanation	Vinegar is a 5-8% solution of acetic acid in water. Used in cooking, cleaning, and pickling.

Part 5: Summary Table of Common Compounds

Common Name	Chemical Formula	IUPAC Name	Found In
Table salt	NaCl	Sodium chloride	Kitchen, food
Baking soda	NaHCO ₃	Sodium hydrogen carbonate	Baking, cleaning
Vinegar	CH ₃ COOH	Acetic acid (dilute)	Cooking, pickling
Household ammonia	NH ₃	Ammonia	Glass cleaners
Sugar	C ₁₂ H ₂₂ O ₁₁	Sucrose	Baking, sweetening
Rubbing alcohol	C ₂ H ₅ OH	Ethanol	Disinfectant, antiseptic
Nail polish remover	(CH ₃) ₂ CO	Acetone	Cosmetics, solvent
Toothpaste fluoride	NaF	Sodium fluoride	Dental care
Muriatic acid	HCl	Hydrochloric acid	Cleaners, masonry
Lime water	Ca(OH) ₂	Calcium hydroxide	Construction, water treatment

Part 6: Tips for Success in Chemical Nomenclature

Tip	Explanation
Identify the compound type	Is it ionic (metal + nonmetal), molecular (nonmetal + nonmetal), or an acid (H + anion)?
Use the periodic table	The position of an element on the periodic table helps predict its ion charge
Memorize common polyatomic ions	Sulfate (SO_4^{2-}), nitrate (NO_3^-), hydroxide (OH^-), carbonate (CO_3^{2-}), etc.
Roman numerals for transition metals	If the metal can have multiple charges, specify with Roman numerals (e.g., iron(II), iron(III))
Prefixes for molecular compounds	Use mono-, di-, tri-, tetra-, penta-, hexa- to indicate the number of atoms
Practice, practice, practice	Naming and writing formulas is a skill that improves with repetition

Key Terms

Term	Definition
IUPAC	International Union of Pure and Applied Chemistry; the organization that standardizes chemical naming
Nomenclature	A system of naming things, especially in science
Ionic compound	A compound formed by the transfer of electrons between a metal and a nonmetal
Molecular (covalent) compound	A compound formed by the sharing of electrons between two or more nonmetals
Polyatomic ion	A charged group of covalently bonded atoms that acts as a single ion
Stock system	A naming system that uses Roman numerals to indicate the charge of a transition metal ion
-ide ending	The suffix added to the name of a nonmetal when it is the second element in an ionic compound
Binary acid	An acid composed of hydrogen and one other nonmetal (e.g., HCl, H ₂ S)
Oxyacid	An acid containing hydrogen, oxygen, and another element (e.g., H ₂ SO ₄ , HNO ₃)

Review Questions

1. What is IUPAC and why is it important in chemistry? Explain why a standardized naming system is necessary for scientists around the world.

2. Name the following compounds using IUPAC rules:

- (a) H_2O
- (b) NaCl
- (c) NH_3
- (d) NaHCO_3
- (e) H_2SO_4
- (f) ZnSO_4
- (g) MgO
- (h) Fe_3O_4
- (i) CaO
- (j) Ca(OH)_2

3. Write the chemical formula for each of the following household compounds:

- (a) Sodium chloride
- (b) Ammonia
- (c) Sodium fluoride
- (d) Hydrochloric acid
- (e) Ethanol
- (f) Sucrose
- (g) Acetone
- (h) Acetic acid

4. Explain the difference between naming an ionic compound (like NaCl) and a molecular compound (like CO_2). Include the rules for each type.

5. For each of the following compounds, state whether it is ionic or molecular (covalent):

- (a) H_2O
- (b) NaCl
- (c) NH_3
- (d) CaO
- (e) MgO

6. Iron can form two different oxides: FeO and Fe_2O_3 . Name each compound using the Stock system (Roman numerals). Explain why Roman numerals are necessary for iron but not for sodium.

7. The compound $\text{Ca}(\text{OH})_2$ is commonly called slaked lime or lime water. Explain the IUPAC name "calcium hydroxide" by identifying the ions present and their charges.

8. Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) and acetic acid (CH_3COOH) are both organic compounds found in the home. State where each is commonly found and one use for each.

9. A student writes the formula for sodium fluoride as NaF_2 . Is this correct? If not, explain the error and write the correct formula.

10. Application question: You find a bottle in the kitchen labeled "acetic acid." Using your knowledge of IUPAC nomenclature and common household substances, explain what this compound is, its chemical formula, and one common use for it in the home